CLASS 8 — SCIENCE · FORMULA SHEET

Chapter 11: **Keeping Time with the Skies** | Quick revision, one page

Earth's motions

- Rotation on its axis: about 24 hours (gives day and night, apparent motion of Sun and stars).
- Revolution around the Sun: about $365\frac{1}{4}$ days (one year; with the tilt of the axis it gives seasons).
- Four extra quarters make 1 day, so a leap year of 366 days comes every 4 years.

Shadows and sundial

- Shadow is longest at sunrise and sunset, shortest at noon.
- Morning shadow points west; evening shadow points east.
- A sundial (gnomon) tells time from the Sun's apparent motion.

Moon

- Moon revolves around the Earth in about 27 days; new moon to new moon is about 29.5 days.
- Phases: new moon, crescent, first quarter, gibbous, full moon, and back.
- Same side of the Moon always faces the Earth. Solar eclipse: new moon; lunar eclipse: full moon.

Calendars and stars

- Solar calendar follows the Sun; lunar calendar follows the Moon phases ($12 \times 29.5 = 354$ days).
- Lunar festivals fall on different dates of the solar calendar each year.
- Pole Star lies almost above the North Pole and appears fixed; other stars seem to circle it.

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